



Short Communication

Unveiling the response of efflux pump genes to delamanid in *Mycobacterium tuberculosis* H37Rv: a transcriptomic analysis

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ABSTRACT

Overexpression of efflux pumps has been associated with drug resistance in *Mycobacterium tuberculosis*. However, their specific role in delamanid resistance remains unclear. This study aimed to identify efflux pump genes influenced by delamanid exposure through transcriptomic analysis.

Cultures of *M. tuberculosis* H37Rv were exposed to delamanid at a sub-inhibitory concentration of $\frac{1}{2}$ MIC for 24 h and transcriptomic sequencing was performed on exposed and unexposed cultures. Differentially expressed genes (DEGs) were annotated using the UniProt database.

Transcriptomic analysis revealed that 23 efflux pump genes were significantly upregulated under delamanid stress, with 17/23 (73.9%) belonging to the ABC family, 4/23 (17.39%) to the RND family and 2/23 (8.69%) to the MFS superfamily. Quantitative real-time PCR (qRT-PCR) was conducted to validate the transcriptomic results for efflux pump genes following exposure of *M. tuberculosis* H37Rv to delamanid at $\frac{1}{2}$ and $\frac{1}{4}$ MIC. qRT-PCR confirmed that 20/23 (86.95%) genes were significantly upregulated at $\frac{1}{2}$ MIC, with 14/20 (70%) of these belonging to the ABC family. In contrast, only 3/23 (13%) efflux pump genes demonstrated upregulation at $\frac{1}{4}$ MIC.

This study identified a set of efflux pump genes significantly upregulated in response to sub-inhibitory concentrations of delamanid and suggested that $\frac{1}{2}$ MIC is a more suitable concentration than $\frac{1}{4}$ MIC for evaluating the expression of efflux pump genes in response to delamanid in *M. tuberculosis*.

1. Introduction

Despite all advances made in treatment and management, tuberculosis (TB) remains one of the main public health problems worldwide (World Health Organization, 2023). The burden of TB is compounded by drug resistant forms of the disease. The current public health issues arise because of emergence of multidrug resistant TB (MDR-TB), extensively drug resistant TB (XDR-TB) and an increase in HIV associated TB. The World Health Organization (WHO) recommends the use of second-line antituberculosis drugs for the treatment of Rifampicin resistant-TB (RR-TB) and MDR-TB (World Health Organization, 2014).

Delamanid is a nitro-dihydro-imidazooxazole previously known as OPC-67683, an anti-mycobacterial drug, that functions by inhibiting the

biosynthesis of keto and methoxy mycolic acid, and received conditional approval for the treatment of adults with MDR-TB, from the European Medicine Agency in 2014 (Szumowski and Lynch, 2015). That same year, the WHO issued interim policy guidance on the use of delamanid for treating RR-TB and MDR-TB (World Health Organization, 2014).

Delamanid exhibits excellent *in vitro* activity against both drug-susceptible and drug-resistant strains of *M. tuberculosis*, with minimum inhibitory concentrations (MICs) ranging from 0.006 to 0.024 mg/L (Matsumoto et al., 2006; Rivers and Mancera, 2008). It is effective against replicating and dormant bacilli (Chen et al., 2017). Although introduced recently, delamanid resistant strains of *M. tuberculosis* have already been observed (Akbari Aghababa et al., 2024; Yoshiyama et al., 2021; Gómez-González et al., 2021). Studies have shown that resistance

Abbreviations: DEGs, Differentially expressed genes; FDR, False discovery rate; MABA, Microplate Alamar Blue Assay; MDR-TB, Multi drug resistant tuberculosis; MIC, Minimum inhibitory concentrations; OADC, Oleic acid, Albumin bovine, Fraction V, Dextrose, Catalase; qRT-PCR, Quantitative real-time PCR; RIF, Rifampicin; RR-TB, Rifampicin resistant-TB; WHO, World Health Organization; XDR-TB, Extensively drug resistant tuberculosis.

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to delamanid can arise due to chromosomal mutations in target genes (Battaglia et al., 2020), including *ddn* which encodes for the deazaflavin (F420)-dependent nitroreductase involved in bioactivation of this pro-drug (Matsumoto et al., 2006; Manjunatha et al., 2006). The other genes involved are *fbiA*, *fbiB*, *fbiC*, and *fgd1*, that encode proteins essential for F420 biosynthesis and the redox cycle. Mutations in *ddn*, *fgd1*, *fbiA*, *fbiB*, and *fbiC* can prevent the metabolism of delamanid and are therefore linked to delamanid resistance (Manjunatha et al., 2006; Fujiwara et al., 2018). Functional assays have suggested that *Rv2073c* contributes to intrinsic resistance by interfering with the inhibition of decaprenylphosphoryl-2-keto- β -D-erythro-pentose reductase (DprE2) through NAD adducts formed during drug activation (Yan et al., 2024). Also, it has been suggested that mutations in *Rv0078*, a transcriptional repressor of *Rv0077*, should be monitored in patients being treated with delamanid or pretomanid (Yan et al., 2024), further demonstrating the diversity of mutations that maybe associated with resistance to delamanid (Rifat et al., 2020).

Though the primary cause of drug resistance in *M. tuberculosis* is the acquisition of mutations in genes that encode drug targets or drug-activating enzymes, *M. tuberculosis* also utilizes various other resistance mechanisms, including overexpression of efflux pumps (Li et al., 2015). Several studies have previously reported overexpression of efflux pumps in drug resistant *M. tuberculosis* isolates (Machado et al., 2012; Li et al., 2015; Garima et al., 2015; Kanji et al., 2017; Narang et al., 2019; Kardan-Yamchi et al., 2019). Inhibition of efflux pumps by efflux pump inhibitors and knockout mutants of efflux pump genes have been associated with improved susceptibility to various drugs (Balganesh et al., 2012). Though efflux pumps have been studied in relation to resistance to newer antituberculosis drugs viz. bedaquiline (Saeed et al., 2022; Madadi-Goli et al., 2024); the involvement of efflux pumps in resistance to delamanid is still unknown.

We, therefore aimed to identify specific efflux pump genes influenced by delamanid exposure through transcriptomic analysis. To the best of our knowledge, this is the first report aiming to contribute valuable insights into the potential role of efflux pumps in delamanid resistance, offering groundwork for future investigations in this field.

2. Materials and methods

2.1. Bacterial strains, growth conditions and reagents

The reference strain *M. tuberculosis* H37Rv was collected from the Department of Microbiology, Vallabhbhai Patel Chest Institute, University of Delhi, Delhi, India and maintained in Middlebrook 7H9 medium (Difco™) supplemented with OADC (oleic acid, albumin bovine, fraction V, dextrose, catalase) (Becton, Dickinson and company sparks) and 2 % glycerol at 37 °C. Delamanid was obtained from MedChemExpress.

2.2. Evaluation of minimum inhibitory concentration (MIC) by Microplate Alamar Blue assay (MABA)

MIC of delamanid was determined for *M. tuberculosis* H37Rv by MABA in 96-well U-bottom plates in triplicates, as described previously with the minor modification of incubating the cultures for 13 days (Franzblau et al., 1998). Medium containing inoculum but no antibiotic was used as growth control. Sterility control, containing medium without inoculum and antibiotic, was also included in the assay. Each concentration of the drug was tested in triplicates.

2.3. In vitro delamanid exposure to *M. tuberculosis* H37Rv

M. tuberculosis H37Rv culture was grown in Middlebrook 7H9 medium (Difco™) till log phase and exposed to delamanid at ½ and ¼ sub-inhibitory concentrations, followed by incubation at 37 °C for 24 hrs. as described previously (Garima et al., 2015). Exposure of ½ MIC of

delamanid was used for both transcriptomic analysis and qRT-PCR, while exposure of ¼ MIC was used for qRT-PCR. All experiments were performed in triplicates. *M. tuberculosis* H37Rv culture unexposed to delamanid was used as control. After incubation, the culture was centrifuged at 8000 RPM for 10 min at room temperature, and the pellet obtained was resuspended in RNA protection buffer (Qiagen, Germany) for RNA isolation and transcriptome sequencing.

2.4. RNA isolation and transcriptome sequencing

RNA from both exposed (sub inhibitory concentration of ½ MIC) and unexposed cultures was isolated with TRIzol (Redcliffe Labs Pvt. Ltd.). The quality of RNA was assessed using Qubit and TapeStation 4150. Samples that passed the quality check parameters were further used for Library Preparation with NEB RNA Library prep kit and subsequently sequenced. Redcliffe Labs analyzed transcriptome samples using Adapter Removal v2.3.2 (Schubert et al., 2016) to filter out the adapters and perform quality trimming at Q30. The clean data were aligned to the *M. tuberculosis* H37Rv reference genome using HISAT2 v2.1.0 (Kim et al., 2015). Gene expression estimation was performed using FeatureCounts v2.0.1 (Liao et al., 2014), and differential gene expression was analyzed using edgeR (Robinson et al., 2010).

2.5. Selection of putative efflux pump genes and primer designing

The DEGs identified in the transcriptomic analysis were listed and annotated using uniprot database (<https://www.uniprot.org/>) (UniProt Consortium, 2019). Further, specific efflux pump genes that were differentially expressed after delamanid exposure were identified using mycobrowser (<https://mycobrowser.epfl.ch/>) and primers for qRT-PCR were designed using Gene Runner Version 3.01 software (<https://www.generunner.com/>) (Gene Runner) (Table S1) for the efflux genes found to be significantly upregulated. Genes were considered significantly upregulated in the transcriptomic analysis if they met the criteria of a false discovery rate (FDR) below 0.05 and a log2 fold change (log2FC) greater than 1 (Shitikov et al., 2022). Homology of the primers was verified using NCBI-blastn tool for microbial genome (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). The primers were synthesized by Imperial Life Sciences Pvt. Ltd. (India).

2.6. Expression analysis of efflux pump genes by qRT-PCR

Quantitative Real time PCR (qRT-PCR) was performed to study the expression of putative or annotated efflux pump genes by using SsoAdvanced Universal SYBR Green Supermix (BIORAD) in CFX96™ Real-time System (C1000 Touch Thermal Cycler, BIORAD) after exposure of a log phase culture of H37Rv to ½ and ¼ MIC of delamanid. The housekeeping genes *sigA* and *rrs* were amplified as internal controls, using published primers (Garima et al., 2015; Narang et al., 2019). Melt curve analysis was performed after each run to confirm the specificity of the primer. Relative quantification was performed to determine overexpression of efflux genes in cultures exposed to delamanid as compared to unexposed cultures.

The data was analyzed using the built-in quantification software. A relative expression equal to one indicated that the expression level was identical to the control, a fold change ≥ 2 was considered as significant overexpression (Narang et al., 2019).

3. Results

3.1. MIC determination and transcriptome analysis

The MIC of delamanid against *M. tuberculosis* H37Rv was determined by the MABA assay as 0.1 mg/L and H37Rv was exposed to a sub-inhibitory concentration of 0.05 mg/L of delamanid for transcriptomic analysis.

Table 1

Graph of the number of differentially expressed genes by functional categories (<https://mycobrowser.epfl.ch>). Total number of genes per functional category listed in Mycobrowser are indicated in parenthesis.

Functional category (Mycobrowser)	No. of genes Upregulated	No. of genes Downregulated
Hypothetical protein (15)	12	3
Stable RNAs (36)	22	14
Information pathways (122)	25	97
Insertion seqs and phages (47)	44	3
PE/PPE (103)	49	54
Regulatory proteins (92)	51	41
Lipid metabolism (119)	57	62
Virulence, detoxification, adaptation (124)	70	54
Cell wall and cell processes (331)	146	185
Intermediary metabolism and respiration (387)	159	228
Conserved hypotheticals (503)	275	228
Total Genes (1879)	910	969

The transcriptome samples were aligned using Hierarchical Indexing for Spliced Alignment of Transcripts2 v2.1.0 (HISAT2 v2.1.0). Gene expression levels were estimated using featureCounts, and statistical analysis of gene expression data was performed using edgeR v3.36.0. DEGs (n = 1879) were identified in delamanid exposed specimen as compared to unexposed culture (Table 1; Fig. 1 a) and 1b)). A total of 910/1879 (48.43 %) genes were upregulated and 969/1879 (51.56 %) genes were downregulated after delamanid exposure. Of the genes that were upregulated, the maximum number of genes belonged to conserved hypotheticals (275/910; 30.21 %), followed by intermediary metabolism and respiration genes (159/910; 17.47 %) and cell wall and cell processes genes (146/910; 16.04 %). Also, 51/910 (5.6 %) regulatory proteins, 49/910 (5.38 %) *pe/ppe* genes were upregulated (Table 1). Similarly, of the down regulated genes, the maximum number belonged to conserved hypotheticals (228/969; 23.52 %), intermediary metabolism and respiration genes (228/969; 23.52 %), and cell wall and cell processes genes (n = 185/969; 19 %) (Table 1).

3.2. Identification of differentially expressed efflux pump genes

Of the cell wall and cell processes genes that were upregulated, 28/146 (19.17 %) were efflux pump genes. These included members of the ABC family (20/28; 71.42 %) including *Rv0194*, *Rv0986*, *Rv0987*,

Rv1218c, *Rv1273c*, *Rv1686c*, *Rv1687c*, *Rv1819c* (*bacA*), *Rv2038c*, *Rv2686c*, *Rv2687c* and others. (Table 2). Members of the MFS and RND families were upregulated to a lesser extent (4/28; 14.28 %). The genes of the MFS family included *Rv0267*, *Rv1250*, *Rv1902c* and *Rv3728*; while genes of the RND family included *Rv0402*, *Rv0507* (*mmpL2*), *Rv1145* (*mmpL13a*) and *Rv2339* (Table 2). Of all the genes, 23 efflux pump genes were found to be significantly upregulated (Table 2), while five genes viz. *Rv1250*, *Rv1273c*, *Rv1819c*, *Rv1902c*, *Rv3663c* had a higher FDR and were excluded from analysis. Therefore, primers were designed for the qRT-PCR analysis for 23 genes (Table S1). Of these 23 efflux pump genes, 17/23 (73.9 %) were part of the ABC family, 4/23 (17.39 %) belonged to the RND family, and 2/23 (8.69 %) were associated with the MFS superfamily. Additionally, 12 efflux genes were downregulated at 1/2MIC exposure of delamanid. Of these, 7/12 (58.33 %) belonged to the ABC family, while three belonged to MFS (3/12; 25 %) and two (2/12; 16.66 %) belonged to RND superfamily (Table 2).

3.3. Expression study of putative efflux pump genes by qRT-PCR

Expression analysis of the 23 efflux pump genes that were significantly upregulated in the transcriptomic study was performed by qRT-PCR to validate the findings of transcriptomic sequencing and to analyze the optimum concentration required for efflux pump gene analysis in relation to delamanid. Hence, an H37Rv culture was exposed to varying sub-inhibitory concentrations ($\frac{1}{2}$ and $\frac{1}{4}$ MIC) of delamanid.

Of the efflux pumps genes analyzed, 20/23 (86.95 %) genes were significantly upregulated by qRT-PCR at $\frac{1}{2}$ sub-inhibitory concentration (Fig. 2). Of these, 14/20 (70 %) belonged to the ABC family, while 4/20 (20 %) belonged to the RND family and 2/20 (10 %) were from the MFS superfamily. Three genes *Rv1218c*, *Rv1686c*, and *Rv1687c*, though they had shown significant upregulation at $\frac{1}{2}$ MIC of delamanid in the transcriptomic analysis, did not show significant upregulation in the qRT-PCR experiment. Only three efflux genes, *Rv0267*, *Rv0987* and *Rv2687c* were significantly upregulated at $\frac{1}{4}$ MIC sub-inhibitory concentration of delamanid, (Fig. 2).

4. Discussion

The approval of bedaquiline, delamanid and pretomanid for the treatment of MDR-TB, has been a key development in managing drug resistant TB (Blair and Scott, 2015; WHO, 2019; WHO, 2022; WHO guide, 2022). Despite their success, resistance to the new drugs has already emerged (Gómez-González et al., 2021; Yang et al., 2018; Mok

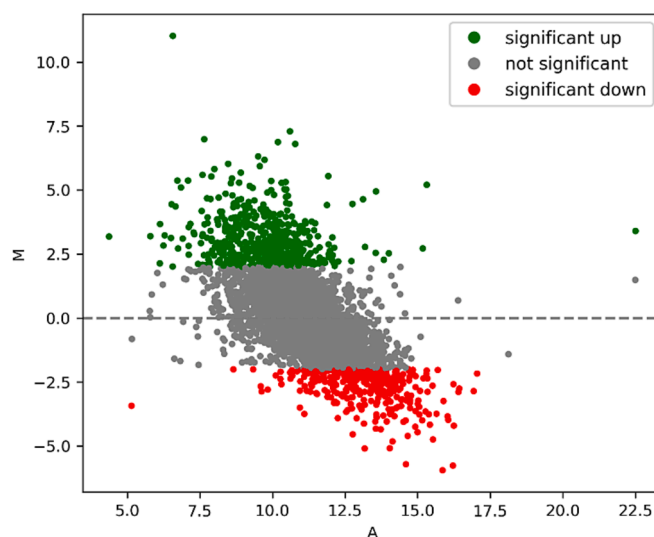


Fig. 1. Differentially expressed genes of *M. tuberculosis* H37Rv in transcriptomic sequencing after exposure to delamanid at $\frac{1}{2}$ MIC (0.025 mg/L) for 24 h. a) MA plot b) heatmap depict the differentially expressed genes in triplicates. Unexposed culture of *M. tuberculosis* H37Rv was used as control.

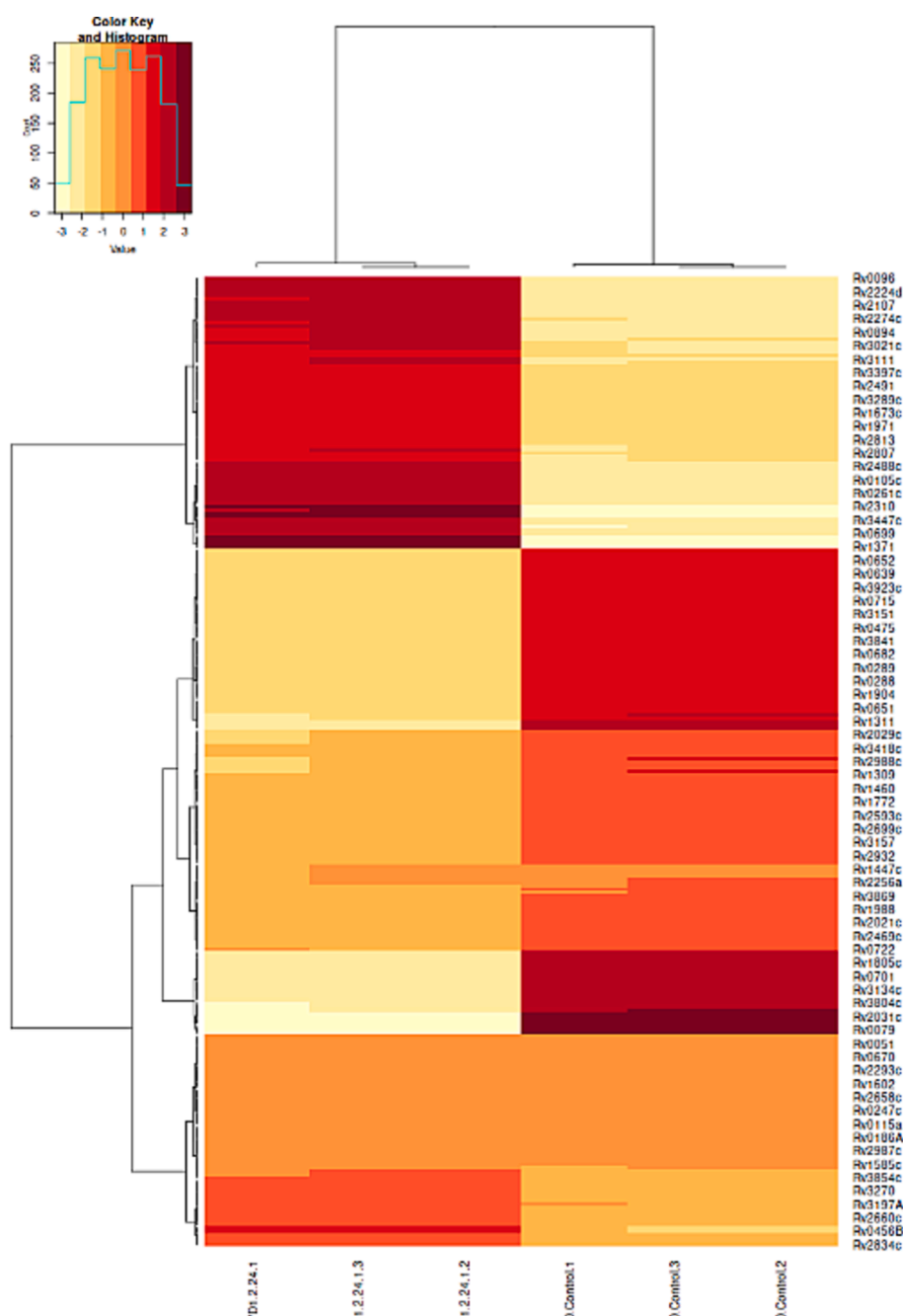


Fig. 1. (continued).

et al., 2023). Delamanid is a group C drug and is advocated for use when drugs in group A (bedaquiline and linezolid) and group B (clofazimine) cannot be used. Resistance to delamanid has been associated with mutations in several genes, including *ddn*, *fgd1*, *fbiA*, *fbiB* and *fbiC* (Manjunatha et al., 2006; Singh et al., 2008; Fujiwara et al., 2018; Nguyen et al., 2020). However, currently, the comprehension of molecular mechanisms involved in delamanid resistance is still inadequate. Moreover, the role of efflux pumps has not yet been explored.

We therefore performed transcriptomic analysis on the laboratory strain H37Rv, after exposure to a subinhibitory concentration of delamanid, and identified significant upregulation in 23 efflux pump genes, with maximum genes (17/23; 73.9 %) belonging to the ABC family of efflux pumps. The results of the transcriptomic analysis were validated by qRT-PCR. While all the genes of the RND and MFS families that had been upregulated in the transcriptomic analysis, showed

increased expression in qRT-PCR, only 14/17 (82.35 %) genes from the ABC family were upregulated after exposure to $\frac{1}{2}$ MIC of delamanid. Moreover, only one gene from the MFS family, and two from the ABC family were upregulated at $\frac{1}{4}$ MIC exposure. Thus, $\frac{1}{2}$ MIC of delamanid was found to be a more effective concentration than $\frac{1}{4}$ MIC for analysis of efflux pump activity.

Of all the efflux genes that were significantly upregulated in both transcriptomic sequencing as well as in qRT-PCR, the ABC family genes *Rv0194*, *Rv2038c*, *Rv2686c*, *Rv2687c*; the RND family *Rv0507*, *Rv1145*; and the MFS family gene *Rv3728* have been studied for efflux pump activity earlier (Kanji et al., 2017; Narang et al., 2019; Kardan-Yamchi et al., 2019; Balganesht et al., 2012; Narang et al., 2017; Wang et al., 2013; Kardan Yamchi et al., 2015; Lee et al., 2021; Pasca et al., 2004; Oh et al., 2017; Hasan et al., 2024; Ghajavand et al., 2019; Gupta et al., 2010; Özgür et al., 2023; Nirmal et al., 2023; El Sahly et al., 2011). In

Table 2

Differentially expressed efflux pump genes in *M. tuberculosis* H37Rv after exposure to Delamanid compared to unexposed culture.

Efflux pumps upregulated				
S. No.	Gene	Efflux Family	logFC	Mycobrowser Product
1	Rv0194	ABC	3.13	Probable transmembrane multidrug efflux pump
2	Rv0267	MFS	2.09	Probable integral membrane nitrite extrusion protein NarU (nitrite facilitator)
3	Rv0402c	RND	2.23	Probable conserved transmembrane transport protein MmpL1
4	Rv0507	RND	1.63	Probable conserved transmembrane transport protein MmpL2
5	Rv0986	ABC	3.52	Probable adhesion component transport ATP-binding protein ABC transporter
6	Rv0987	ABC	2.47	Probable adhesion component transport transmembrane protein ABC transporter
7	Rv1145	RND	1.86	Probable conserved transmembrane transport protein MmpL13a
8	Rv1218c	ABC	1.75	Probable tetracycline-transport ATP-binding protein ABC transporter
9	Rv1686c	ABC	1.68	Probable conserved integral membrane protein ABC transporter
10	Rv1687c	ABC	1.85	Probable conserved ATP-binding protein ABC transporter
11	Rv2038c	ABC	3.62	Probable sugar-transport ATP-binding protein ABC transporter
12	Rv2039c	ABC	3.57	Probable sugar-transport integral membrane protein ABC transporter
13	Rv2040c	ABC	4.15	Probable sugar-transport integral membrane protein ABC transporter
14	Rv2316	ABC	2.49	Probable sugar-transport integral membrane protein ABC transporter UspA
15	Rv2317	ABC	4.23	Probable sugar-transport integral membrane protein ABC transporter UspB
16	Rv2339	RND	2.20	Probable conserved transmembrane transport protein MmpL9
17	Rv2686c	ABC	2.11	Antibiotic-transport integral membrane leucine and alanine and valine rich protein ABC transporter
18	Rv2687c	ABC	2.91	Antibiotic-transport integral membrane leucine and valine rich protein ABC transporter
19	Rv2834c	ABC	2.59	Probable Sn-glycerol-3-phosphate transport integral membrane protein ABC transporter UgpE
20	Rv2835c	ABC	2.92	Probable Sn-glycerol-3-phosphate transport integral membrane protein ABC transporter UgpA
21	Rv3664c	ABC	2.71	Probable dipeptide-transport integral membrane protein ABC transporter DppC
22	Rv3665c	ABC	2.33	Probable dipeptide-transport integral membrane protein ABC transporter DppB
23	Rv3728	MFS	2.47	Probable conserved two-domain membrane protein
Efflux pumps downregulated				
S. No.	Gene	Efflux family	logFC	Mycobrowser Product
1	Rv3197	ABC	−1.22	Probable conserved ATP-binding protein ABC transporter
2	Rv0655	ABC	−2.28	Possible ribonucleotide-transport ATP-binding protein ABC transporter Mkl
3	Rv0935	ABC	−2.15	Phosphate-transport integral membrane ABC transporter PstC1
4	Rv2477c	ABC	−2.09	Probable macrolide-transport ATP-binding protein ABC transporter
5	Rv2564	ABC	−1.31	Probable glutamine-transport ATP-binding protein ABC transporter GlnQ
6	Rv2326c	ABC	−1.27	Possible transmembrane ATP-binding protein ABC transporter
7	Rv0073	ABC	−1.23	Probable glutamine-transport ATP-binding protein ABC transporter

Table 2 (continued)

Efflux pumps upregulated				
S. No.	Gene	Efflux Family	logFC	Mycobrowser Product
8	Rv1410c	MFS	−2.48	Aminoglycosides/tetracycline-transport integral membrane protein
9	Rv2846c	MFS	−2.16	Possible integral membrane efflux protein EfpA
10	Rv3823c	RND	−1.07	Conserved integral membrane transport protein MmpL8
11	Rv0206c	RND	−2.19	Possible conserved transmembrane transport protein MmpL3

previous studies in our laboratory, Rv0194 of the ABC family, was associated with resistance to isoniazid (Narang et al., 2017), but no significant correlation was observed with rifampicin resistance (Narang et al., 2019). In a study from Pakistan, no significant difference was seen in the expression of Rv0194 in XDR and drug susceptible clinical isolates of *M. tuberculosis* (Kanji et al., 2017). Rv2038c is thought to contribute to kanamycin resistance (Lee et al., 2021). In a study by Pasca et al., the *M. tuberculosis* Rv2686c-Rv2687c-Rv2688c operon was shown to confer resistance to fluoroquinolones when expressed in *M. smegmatis* (Kardan Yamchi et al., 2015). Expression of this operon resulted in a significant increase in resistance to ciprofloxacin, and to a lesser extent, norfloxacin, moxifloxacin, and sparflaxacin. The observed resistance was mitigated in the presence of known efflux pump inhibitors such as reserpine, carbonyl cyanide m-chlorophenylhydrazone (CCCP), and verapamil, indicating the involvement of an energy-dependent efflux mechanism. (Pasca et al., 2004). However, Rv2686c and Rv2687c were not found to be associated with fluoroquinolone resistance in another study (Oh et al., 2017). Surprisingly, though the ABC family genes Rv1218c, Rv1686c and Rv1687c had shown upregulation in RNA sequencing, they were not upregulated in the qRT-PCR experiment, though, there is evidence that Rv1218c has a significant function in facilitating efflux of various antibiotics including rifampicin (Wang et al., 2013) and this efflux pump has been seen to be overexpressed in MDR and XDR isolates of *M. tuberculosis* (Kardan Yamchi et al., 2015), while Rv1686c and Rv1687c have recently been associated with kanamycin resistance (Lee et al., 2021).

Amongst the genes of the RND superfamily upregulated in the present study, Rv0507 has been earlier associated with resistance to isoniazid (Narang et al., 2017; Hasan et al., 2024), although no link has been observed with rifampicin resistance (Narang et al., 2019). Another gene of the RND family, Rv1145 that was significantly upregulated in our study, has also been associated with resistance to isoniazid and rifampicin (Ghajavand et al., 2019).

Similarly, the MFS superfamily gene Rv3728, has been observed to be overexpressed in MDR-TB isolates, after exposure to antituberculosis drugs (Gupta et al., 2010) and has also been linked to kanamycin resistance (Lee et al., 2021).

However, several (13/23; 56.52 %) genes encoding efflux pumps, though found to be significantly upregulated in our study, had no earlier reports in relation to their efflux pump activity. These included the ABC family genes Rv0986, Rv0987, Rv2039c, Rv2040c, Rv2316, Rv2317, Rv2834c, Rv2835c, Rv3664c, Rv3665c; the RND superfamily gene Rv0402c, Rv2339; and the MFS superfamily gene Rv0267.

Importantly, none of the 20/23 (86.95 %) genes significantly upregulated in the present study (Fig. 2), in response to delamanid exposure, have been studied in relation to this drug. Analysis of these efflux genes in relation to delamanid resistance in clinical isolates of *M. tuberculosis* would pave the path for future studies on alternative mechanisms of delamanid resistance.

In conclusion, the rapid evolution of delamanid resistance in MDR-TB and XDR-TB, even though the drug has been introduced recently, is alarming (Bloembergen et al., 2015). Hence, despite mutations in several genes viz *fbIA*, *fbIB*, *fbIC*, *ddn* and *fgd1* having been implicated in drug

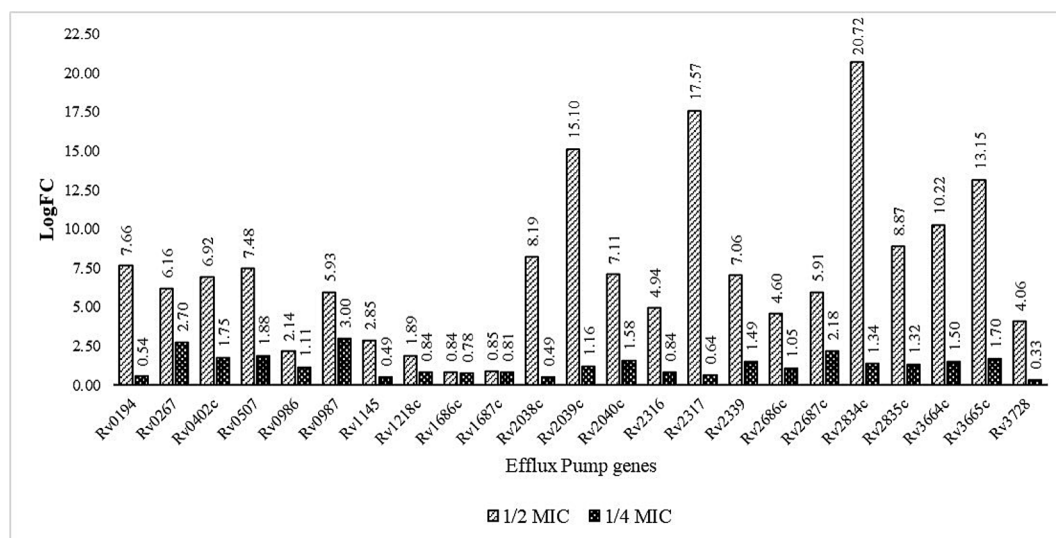


Fig. 2. Relative expression of 23 efflux genes, that were upregulated in the transcriptomic analysis, in *M. tuberculosis* H37Rv exposed to $\frac{1}{2}$ (0.025 mg/L) and $\frac{1}{4}$ (0.0125 mg/L) MIC concentrations of delamanid. The delamanid exposed culture was analyzed in comparison with unexposed control. *Rv1218c*, *Rv1686c* and *Rv2038c* did not exhibit significant upregulation.

resistance, alternate mechanisms such as efflux need to be explored. The present study has, for the first time, identified a group of efflux pump genes that may be analyzed further for their role in delamanid resistance. In addition, we determined $\frac{1}{2}$ MIC of delamanid to be more appropriate than $\frac{1}{4}$ MIC for evaluating efflux pump activity.

Data deposition

Raw data files are available in Gene Expression Omnibus (Geo) in NCBI under accession numbers GSE287913 (study), and GSM8754368; GSM8754369; GSM8754370; GSM8754371; GSM8754372; GSM8754373 (dataset).

CRedit authorship contribution statement

Varsha Chauhan: Writing – original draft, Methodology, Investigation, Data curation. **Anupriya Singh:** Methodology. **Anmol Guleria:** Methodology. **Swati Kathait:** Methodology. **Kamal Shrivastava:** Methodology. **Chanchal Kumar:** Methodology. **Shalini Suman:** Methodology. **Sanjay Kumar:** Supervision, Formal analysis. **Mandira Varma-Basil:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gene.2025.149661>.

Data availability

We have shared the data in the manuscript.

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